Computer Systems – Final Report

Group: Clinic IT Team

Members: (Add Names & Student IDs)

Introduction

This report evaluates the existing IT infrastructure of the City Health Clinic, analyses parallel computing needs, proposes a new network design, and delivers a complete budget estimation. The goal is to provide a reliable, scalable, and high■performance system suitable for four departments: Emergency, Outpatient, Radiology, and Administration.

PART A — Benchmarking & Performance Analysis

Your personal system Novabench results:

CPU: 1317

GPU: 10

Memory: 294

Storage: 162

Device Evaluation Summary

Modern devices such as:

- HP Envy (11th Gen)
- ASUS ZenBook 14
- MacBook Pro (assumed M1 model)

are suitable for continued use due to strong CPU and SSD performance.

Older devices such as:

- HP Envy (7th Gen)
- MSI 9th Gen
- Lenovo Ideapad

show significant performance bottlenecks and should be replaced.

Simulated CPU Task (1 million cycles)

Execution time approximation:

Time = cycles / clock_speed

Modern CPUs (~3.0 GHz) finish ~0.00033 ms

Older CPUs (~1.9 GHz) finish ~0.00052 ms

Difference justifies upgrading outdated systems.

Graphs (not included in PDF, but provided in written content):

• CPU comparison bar chart

• RAM bar chart

• Disk speed bar chart

PART B — Parallel Computing Needs

Parallel computing improves performance by dividing tasks across multiple processors or cores. For the clinic:

Emergency Department:

Real■time processing, fast retrieval.

Radiology Department:

Large imaging files benefit heavily from GPU acceleration, shared memory parallelism.

Administration & Outpatient:

Parallel databases allow faster record access and appointment handling.

Limitations:

• Not all tasks can be parallelised

• Synchronisation overhead

• Hardware cost increases

Framework Evaluation:

• CUDA — ideal for radiology GPU tasks

• OpenMP — good for shared memory, multi■core CPUs

• MPI — suitable for distributed systems

Business Translation:

• Faster patient service

• Reduced waiting time

• Higher staff efficiency

• Better data processing reliability

PART C — Network Design

Department VLANs:

Emergency — VLAN 10

Outpatient — VLAN 20

Radiology — VLAN 30

Administration — VLAN 40

Network Devices:

• Core router

• Layer■3 switch

• Department switches

• Access Points

• Firewall

• Servers (Database, Imaging, Backup)

Security:

• Access Control Lists

• VLAN segmentation

• Encrypted wireless

• Daily backups

• Secure DNS/DHCP

Scalability:

• Additional floors or devices can be added by expanding VLAN addressing

• Modular switches support more ports

PART D — Budget Estimation

Estimated Costs (sample):

• New PCs (6 units) — £4,200

• Servers (2 units) — £6,000

• Switches & Router — £2,500

• Access Points — £600

• Cabling & Installation — £1,200

• Software licenses — £1,000

Total Estimated Cost: £15,500

Cost■Reduction Strategy:

• Use cloud services for storage

• Replace only outdated devices

• Bulk hardware purchase

Contingency Budget:

If funding is cut, minimum■viable setup = £9,000.

Conclusion

This assessment provides a robust plan for evaluating hardware, implementing parallel computing, designing a secure network, and estimating realistic costs. The proposed infrastructure ensures reliability, performance, and long■term scalability for the City Health Clinic.